Sanchaai Mathiyarasan

437-436-4126 | sanchaai.mathi@gmail.com | sm0.ca | github.com/sm0ca | linkedin.com/in/sanchaai

Education

University of Toronto – Honours BSc. Computer Science Specialist, Statistics Minor — Sep 2022 – Apr 2026

- **CGPA: 3.93/4.0**; Dean's List Scholar; Ambassador for the Department of Computer Science; UofT Linux User Group Executive
- **Relevant Coursework**: Software Design, Foundations of Computer Science I & II, An Introduction to Statistical Reasoning and Data Science, Calculus with Proofs

Experience

SUDS Research Scholar – Data Sciences Institute — May 2023 – Aug 2023

- Selected as one of **35** students from a highly competitive pool of **200+** applicants from across Canada to participate in the 4-month Summer Undergraduate Data Science (SUDS) Research Program with funding
- Analyzed and improved the performance of a network community detection algorithm by using mathematical optimization, under the supervision of Prof. Samin Aref (Dept. of Mechanical and Industrial Engineering)
- Generated **600+** network graphs using a newly developed model implemented in **Julia** to compare, evaluate, and create visualizations based on the performance of **20** heuristic community detection algorithms in **Python**

Administrative Assistant – Royal Cartridges — Jun 2022 – Dec 2022

- Improved the efficiency of completing post-transaction procedures by up to **30%** by using **Python** GUI-based automation and image recognition libraries to automate tedious and repetitive tasks
- Constructed spreadsheets to compute and keep track of monthly sales and modified existing sheets for ease-of-use
- Increased the speed, effectiveness, and reliability of previous implementations of the automation scripts
- Participated in project meetings with executive staff members to foster company growth

Projects

UT-Tabler – github.com/sm0ca/ut-tabler — Mar 2023 – Apr 2023

- Developed a program to enable UofT students to specify courses as well as schedule-preferences (i.e., minimizing distance between buildings, or avoiding morning/evening classes) to automatically generate up to **15** optimal timetables
- Leveraged **Python**'s Selenium and BS4 libraries to scrape, parse, and store details pertaining to **4500+** course sections
- Generated timetables by accumulating course sections in a tree data structure, and ranking each based on computed scores
- Collaborated with teammates both remotely and in-person to complete and document the project while meeting deadlines

Moview – github.com/sm0ca/moview — May 2022 – Jun 2022

- Designed and implemented an application to enable users to search for and view movie details (i.e., ratings, runtime)
- Gathered info using HTTP requests, parsed the resulting JSON data, and displaying results in a GUI interface
- Implemented genre-based recommendations, allowing users to cycle through and click posters to find similar films
- Enabled users to maintain separate accounts/watchlists by employing a locally-stored database using text files

Chapter Check – github.com/sm0ca/chapter-check — May 2021 – Jun 2021

- Created software to help users keep track of their progress within manga (Japanese comic) series
- Used requests library and Google Programmable Search Engine to search for and scrape up-to-date information with respect to the manga series that have been specified by the user
- Compared website data against information parsed from text records associated with the logged-in account to notify the user of any new chapter releases for series in their personal manga list

Skills

Languages Python, Java, HTML/CSS, C++ (Arduino), SQL, R

Libraries Flask, Selenium, BS4, Tkinter, NumPy, Plotly, PyAutoGUI, Pillow, requests, pandas, pytest, tidyverse

Dev Tools Git, GitHub, Android Studio, VS Code, Vim, PyCharm, Jupyter, Linux, Bash, OOP, Agile Scrum, UML, MS Office